

(3.2.) $\langle f, g \rangle = \int_1^5 f(x)g(x) dx$, $U = \text{span}\{1, x, x^2\}$

a) $\mu_1 = 1$
 $\|\mu_1\| = \sqrt{\langle 1, 1 \rangle} = \sqrt{\int_1^5 1 \cdot 1 dx} = \sqrt{4} = 2$ $e_1 = \frac{\mu_1}{\|\mu_1\|} = \frac{1}{2}$ = první vektor

$\mu_2 = x - \langle x, e_1 \rangle e_1 = x - \langle x, \frac{1}{2} \rangle \frac{1}{2} = x - 6 \cdot \frac{1}{2} = x - 3$

$\langle x, \frac{1}{2} \rangle = \int_1^5 x \cdot \frac{1}{2} dx = 6$

$\|\mu_2\| = \sqrt{\langle x-3, x-3 \rangle} = \sqrt{\int_1^5 (x-3)^2 dx} = \frac{4}{\sqrt{3}}$

$e_2 = \frac{\mu_2}{\|\mu_2\|} = \frac{x-3}{\frac{4}{\sqrt{3}}} = \frac{\sqrt{3}}{4}(x-3)$ = druhý vektor

$\mu_3 = x^2 - \langle x^2, e_1 \rangle e_1 - \langle x^2, e_2 \rangle e_2$

$\langle x^2, e_1 \rangle = \langle x^2, \frac{1}{2} \rangle = \int_1^5 x^2 \cdot \frac{1}{2} dx = \frac{62}{3}$

$\langle x^2, e_2 \rangle = \langle x^2, \frac{\sqrt{3}}{4}(x-3) \rangle = \int_1^5 x^2 \frac{\sqrt{3}}{4}(x-3) dx = \frac{\sqrt{3}}{4} \cdot 32 = 8\sqrt{3}$

$\mu_3 = x^2 - \frac{62}{3} \cdot \frac{1}{2} - 8\sqrt{3} \cdot \frac{\sqrt{3}}{4}(x-3) = x^2 - \frac{31}{3} - 6(x-3) = x^2 - 6x + 18 - \frac{31}{3}$

$= x^2 - 6x + \frac{23}{3} \Rightarrow \mu_3 = 3x^2 - 18x + 23$

↳ vynásobením vektoru 3-krát na ortogonalitu normování, ale bude se s tím lépe počítat

$\|\mu_3\| = \sqrt{\int_1^5 (3x^2 - 18x + 23)^2 dx} = \frac{16}{\sqrt{5}}$

$e_3 = \frac{\mu_3}{\|\mu_3\|} = \frac{\sqrt{5}}{16}(3x^2 - 18x + 23)$ = třetí vektor

ortonormální báze je tedy $\left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{\sqrt{5}}{16}(3x^2 - 18x + 23) \right)$

b) projeksija $\sin x$ na ON bazu:

$$\text{proj}_B \sin x = \langle \sin x, l_1 \rangle l_1 + \langle \sin x, l_2 \rangle l_2 + \langle \sin x, l_3 \rangle l_3$$

$$\langle \sin x, l_1 \rangle = \int_1^5 \frac{1}{2} \sin x \, dx \doteq 0,128$$

$$\langle \sin x, l_2 \rangle = \int_1^5 \sin(x) \cdot \frac{\sqrt{3}}{4} (x-3) \, dx = -1,493$$

$$\langle \sin x, l_3 \rangle = \int_1^5 \sin(x) \cdot \frac{\sqrt{5}}{16} (3x^2 - 18x + 23) \, dx = -0,125$$

$$\text{proj}_B \sin x \doteq 0,128 \cdot \frac{1}{2} - 1,493 \cdot \frac{\sqrt{3}}{4} (x-3) - 0,125 \cdot \frac{\sqrt{5}}{16} (3x^2 - 18x + 23)$$

$$\doteq 0,064 - 0,646x + 1,438 - 0,0524x^2 + 0,314x - 0,402$$

$$\doteq -0,0524x^2 - 0,332x + 1,600$$

$$\underline{a = -0,052 \quad b = -0,332 \quad c = 1,600}$$

$$c) \|\sin x - (-0,0524x^2 - 0,332x + 1,600)\| = \sqrt{\int_1^5 (\sin x - (-11-)) \, dx}$$

$$\underline{\doteq 0,319} = \text{velikost chyby aproximácie}$$